

Comparing principally imagination and interaction versions of a play anywhere mobile AR location-based story

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ABSTRACT

Augmented Reality (AR) allows virtual elements to be overlaid on the real world, providing new opportunities for location-based storytelling, by offering blended environments to more closely resemble those in a story. However, there is limited research considering how different interaction opportunities with such augmented surroundings, affect user engagement and immersion in such a story. A mobile AR app, Map Story 2, was developed to investigate this, offering a guided story-walk around a user's chosen location, either interacting with overlaid virtual objects to progress the story, or being asked to imagine the same events playing out at each augmented location.

Index Terms: Human-centered computing—Human computer interaction (HCI)—Interaction paradigms—Mixed / augmented reality;

1 INTRODUCTION

Mobile AR has the potential to offer unique interaction and location-based storytelling opportunities through a mobile phone. Murray suggested immersion, agency and transformation were key factors for any new storytelling medium, for a the user to feel authentically engaged [9]. Whilst literary narratives can be read anywhere, encouraging a reader to use their imagination to fill in missing details, expanding on Kendall Walton's ideas it has been suggested that virtual objects might act as props, to encourage a user to engage in make-believe play with them, that might similarly support the questioning of the people and events depicted [7]. Virtual content also offers interaction opportunities as found in video games, providing an alternate way to immerse a user in an experience. Immersion may occur from a desire to be transported into the story events, or perceptual immersion from recognising a similarity with the user's surroundings and those depicted, that Reid termed *magic moments* in location-based games [8] [10].

Given limited prior research as to how different opportunities with an augmented scene might affect users' engagement and immersion in a story, a new app, Map Story 2 was created. It used mobile AR, GPS, and local place data, to offer an AR story experience at a user's chosen location. A mixed methods in the wild study was adopted to trial the app, principally imagination and interaction versions offering two approaches to progress the story through relating to each AR scene. A within-subjects approach further tested whether incorporating real world place names into the story had an effect on users' engagement, with users demonstrating a preference for these names when incorporated. The two versions of the app offered similar engagement, though with greater variation in the imagination version, suggesting this version offered greater challenges for some users. Data was collected through the app via existing questionnaires, open feedback, as well as post-interviews with several participants.

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2 RELATED WORK

The study of mixed reality location-based experiences is challenging since they incorporate multiple related elements [11]. Benford's framework considers participants' *trajectories*, that will commonly vary from the designer's idealised path [1]. Map Story 2 examined previous related research, that suggested how to deal with issues like limited GPS accuracy, the need for on-boarding, and disorientation caused by virtual objects shifting relative to the real world. The app uses *Simultaneous Localisation and Mapping* (SLAM), to offer markerless AR image tracking, without needing physical markers [6]. Challenges arise in transposing such an experience to various locations whilst maintaining story coherence, and varying the amount of walking or atmosphere each location brings [4]. Using AR follows earlier experiences made for the heritage sector and the video narrative, *iLand*, which suggested greater immersion and mental imagery occurs when a story is viewed at a location with a similar atmosphere [5]. Further issues arise from self consciousness taking part in such an experience in public, that may cause users to disengage [3]. Salen and Zimmerman's *immersive fallacy* suggests users crave immersion, even at the expense of some realism, as long as the barriers to it are not too great [12]. Deterding discusses the role imagination plays in video games through theming, the choice of story, the framing adopted and opportunities for role-play [2]. Map Story 2 varies the framing and user opportunities to investigate their effect, with users asked to imagine events occurring at each AR location, or interacting with them, tied to the same events.

2.1 The Map Story 2 App

Map Story 2 was made for Android and iOS phones capable of supporting AR through the camera. Users became the main protagonist in the story, piecing together a forgotten memory by visiting six sites near their starting location. The sites were selected according to their prominence on the Google Places API, the shortest walk calculated between them. On arriving at each site, GPS detection would progress the story automatically, tools provided to help place virtual objects according to their described story locations. The app would then progress differently based on whether the user had randomly been assigned to the imagination or interaction version of the app. In the imagination version users would be asked to imagine a series of events playing out in relation to the AR scene, through a sequence of prompts, this lasting around 30 seconds. In the interaction version, users would interact with the virtual content using the phone's accelerometer and/or touchscreen, tied to the same events. As an example, users placed a 3D model of a bench, and in the imagination condition were asked to imagine searching for and ultimately finding a lost letter, prompts guiding their imagined search. In the interaction version, the user would be asked to actively search the bench, the letter appearing tied to movement of the phone's accelerometer.

The story would gradually be revealed after visiting each site and completing the interaction there, with an option to have the text read out through *Text To Speech* (TTS). Three out of the six sites visited also randomly incorporated the name of the real world location into the story text to examine whether this had a significant effect on users' preference for these sites. After visiting all six sites, users

placed a final AR scene, corresponding to the location where they selected the fictional memory to have taken place. This final AR scene included elements of the virtual models from the six previous AR locations, and tied the events together to offer an ending to the story. The app would take around 20-25 minutes to complete, taking place within 250 metres of the user's starting location.

3 MAP STORY 2 STUDY

Map Story 2 provided a location-based story for use almost anywhere, a welcome opportunity during COVID restrictions. A study with the app investigated the following research questions:

1. Did users report a significantly different experience between the imagination and interaction versions of the app?
2. Could any differences in experience be related to aspects of the users taking part?
3. Was there a preference for the sections of the story that were personalised by referencing a user's current real world location?

Data was checked and retained from 58 participants across both versions of the app (30 male, 26 female, 2 other gender identity), with some users recruited using an online study recruitment site. Due to its anonymity, the 12 participants willing to take part in a follow-up interview came from other recruitment channels. All participants first completed the *Immersive Tendencies Questionnaire* (ITQ), previously validated for VR, as a first test whether this might correlate with different users' immersion reported. Users also rated statements on three occasions whilst taking part, to measure how their engagement in the story and experience varied. On finishing, four sections taken from existing questionnaires were also completed. These were the transportation and emotional engagement sections of the *Story World Absorption* (SW) questionnaire, the mental immersion (engagement) factor of the *Temple Presence Inventory* (TPI), and the usability section of the *Augmented Reality Immersion* (ARI) questionnaire.

4 RESULTS AND DISCUSSION

Accounting for the data collected showing departures from a normal distribution, there were limited differences in the ratings of the two versions of the app, apart from evidence of a larger spread in emotional engagement in the imagination version ($F = 5.1, p = 0.03$) using a Levene test. A paired Wilcoxon test also suggested each users immersion in the story decreased during the experience in the interaction version, compared to the imagination one ($r = 0.23, p = 0.05$). This suggests areas for further exploration, where some users may have struggled to use their imagination in a potentially distracting public space, compared to the offer of a clearly defined interaction. A balance between the two versions or variety of tasks might prove most engaging, given some users of the imagination version commented on attempts to interact, "I tried to peer in to the bin to see if there was a clue in there". Ratings across the four measures collected are shown in Figure 1. In both versions users tended to rate the transportation factor lower than other measurements, suggesting the experience generated a limited sense of presence. No relationship was found between any user demographics including the immersive tendency scores, and the four measures collected. However, barriers to deeper immersion for different users was suggested by feedback, such as in terms of a varying preference for walking. Across both versions users showed a preference for the inclusion of local place names using a Chi-squared test ($\chi^2 = 8.6, p = 0.01$), with some users suggesting that not all sites visited were "important or exciting", and this could have enticed some to walk further. Trying to match the description of where objects were located in the story proved favourable, "knowing that the bike needed to be locked to railings gave an interesting extra component to placing it", with the AR positioning tools to achieve this being well received.

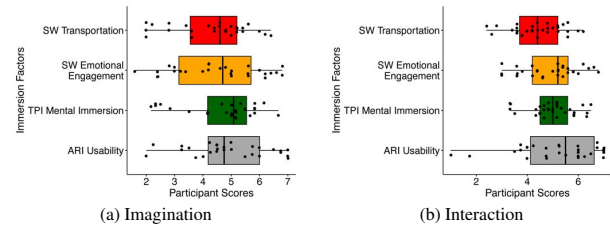


Figure 1: The four collected measure rated by all participants across the imagination, (a) and interaction, (b) versions of Map Story 2.

4.1 Conclusions and Further Work

Map Story 2 offered an AR story that could be played anywhere, at a time when many related experiences were closed. COVID restrictions influenced the study approach adopted, the sample size as well as risking some confounding effects from different handsets and user environments. However, the study provided a first test as to different user responses to using AR scenes as props to guide the imagination, compared to interacting with virtual objects directly. The lack of significant differences between the two versions suggest both can offer an engaging user experience. Further work is required to combine both opportunities most effectively, to offer an increased sense of presence in the story world, whilst accounting for different preferences in relation to the amount of walking, and self consciousness from interacting with virtual objects in public. Additional research directions come from personalisation, to test whether the preference for local place names may just be a fleeting novelty, as well as the effect of offering greater user agency to affect the story direction.

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